

Favourite-longshot bias, stratified

Is the textbook regularity real on our tape — where, how big, and bigger than what it costs?

The literature says recreational flow overpays for longshots and underpays favourites. We measured it on 16 sealed days of the whole exchange, one cell per family $\times$ side $\times$ price bucket $\times$ time-to-close band, with the market as the sample unit.

133 / 560

cells where the deviation beats the round-trip cost, with $n \geq 30$ and a CI clear of zero

6 shapes

market families measured — and six different pictures. There is no single "longshot bias"

2 FAIL

families whose 45–55¢ null cell fails: econ and politics. That is not FLB, it is directional YES optimism

111.7M

trade prints joined to settlement truth (82.3% of the tape)

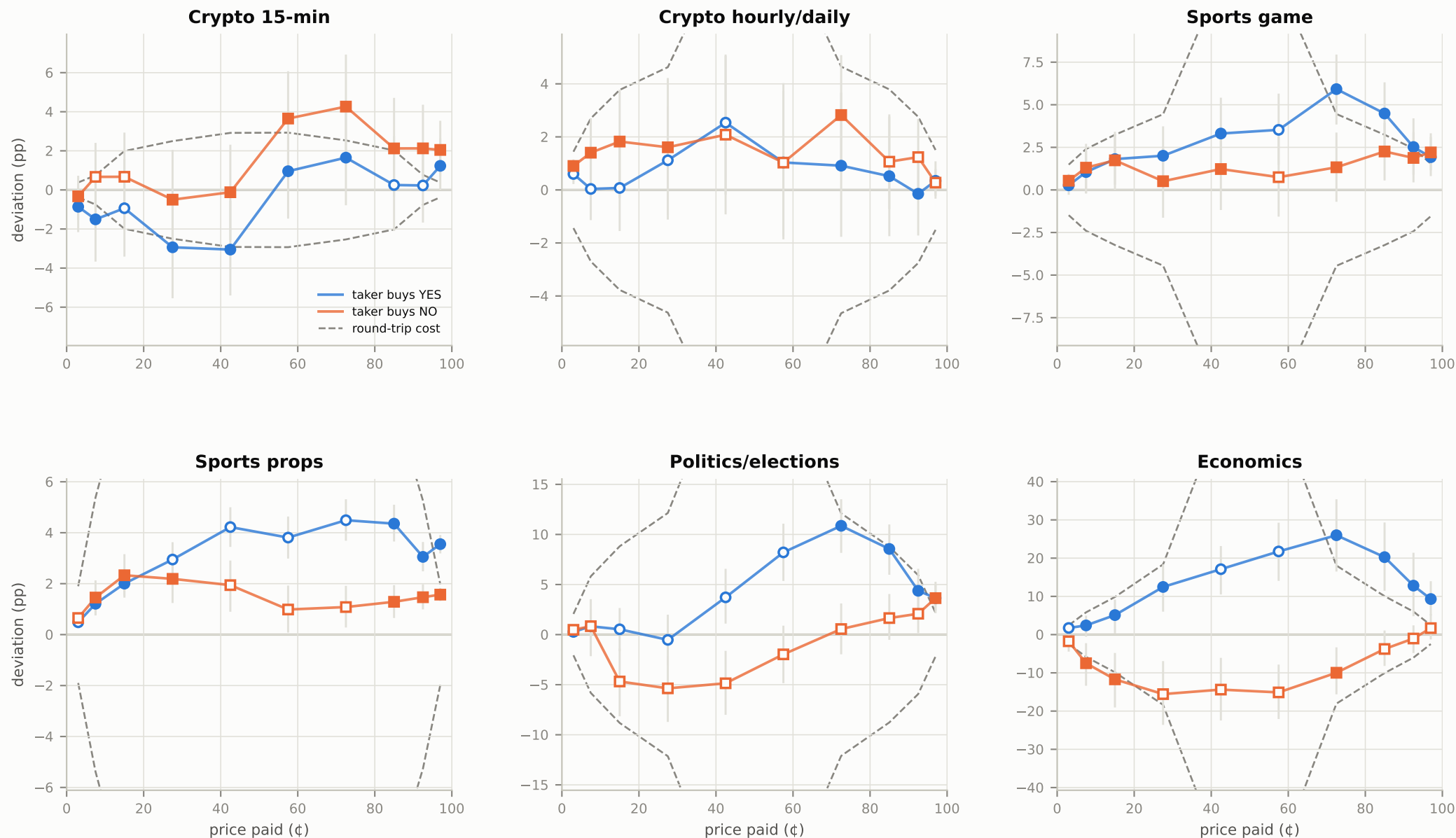
What the pictures say.

- ① Crypto hourly is the textbook case — longshots over-priced, favourites under-priced, and the deviation shrinks as the market approaches close ($3.35 \rightarrow 1.23\text{pp}$). That is FLB behaving exactly as written.
- ② Sports near the close is a different animal. The deviation grows into the final minutes instead of shrinking ($4.91 \rightarrow 6.84\text{pp}$ on single games, $3.38 \rightarrow 9.20$ on props). Diagnosed as end-of-game lottery behaviour, not a broken anchor—settlement is never stamped before close anywhere in the warehouse.
- ③ Economics and politics fail the null-hypothesis cell. At 45–55¢, where a price-bucket story predicts no deviation at all, econ shows YES $+18.8\text{pp}$ / NO -14.3pp . A price-symmetric bias cannot do that; a directional over-pricing of YES can. Those two families must be read after netting the directional component.
- ④ Complementarity holds, which is the check that the measurement is not an artifact: of the 90 pairs where both sides deviate by $\geq 2\text{pp}$, 84 (93.3%) point in opposite directions, as two sides of the same contract must.

F1 · The shape of the bias, family by family

bias by price bucket

Deviation = price paid – realized win rate, per price bucket, pooled over time bands and weighted by markets. Above zero the taker overpaid. Filled markers cleared the round-trip cost (dashed); hollow ones did not.

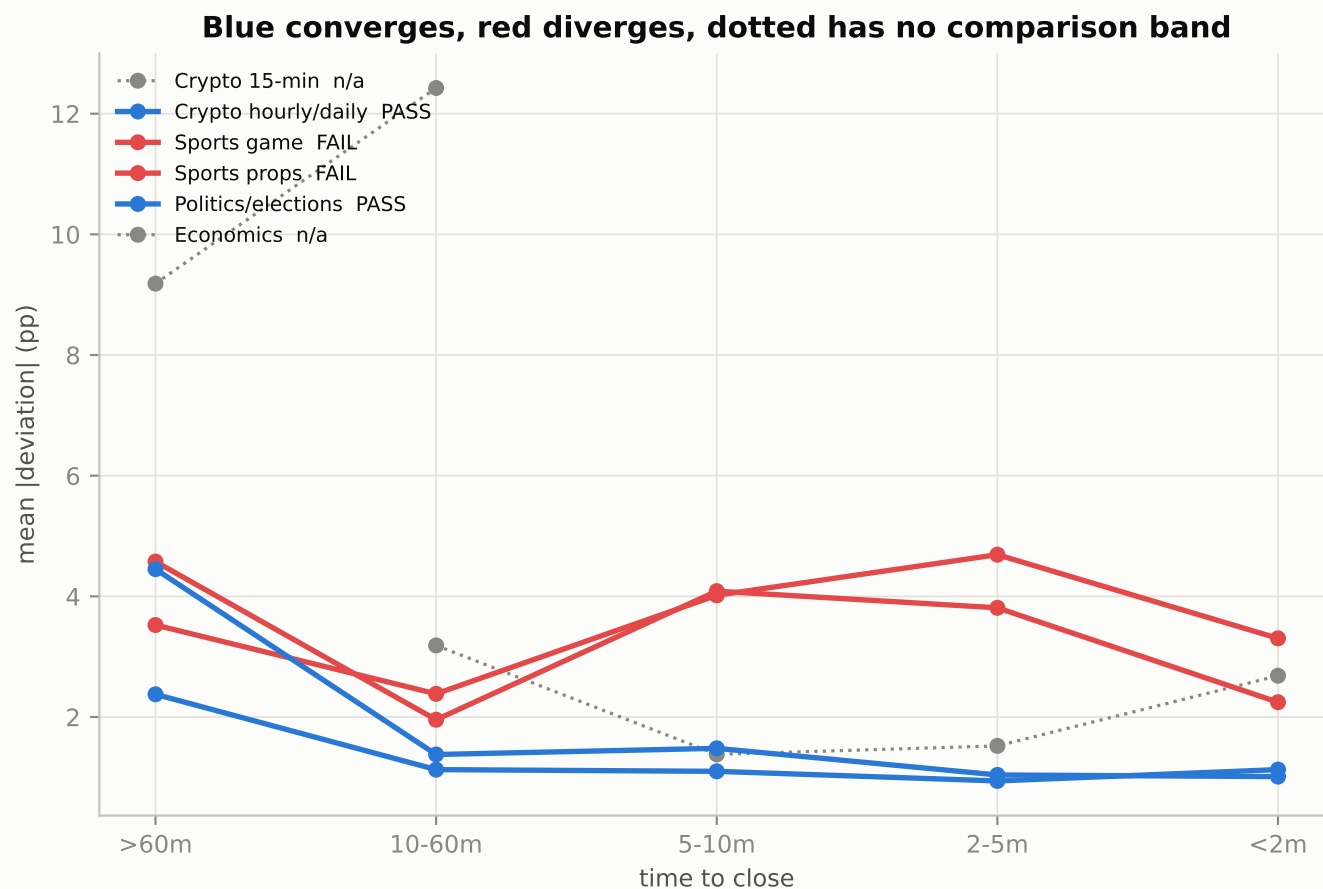


Error bars are 95% cluster bootstrap over markets; the dashed cost lines run off the panel in the thin families, which is the finding, not a plotting accident. Classic FLB reads as points above zero on the left and below zero on the right— crypto hourly and, more weakly, crypto 15-min show it. Sports props sit above zero at BOTH ends, and carry the largest statistical mass in the study (95-99¢ YES +5.86pp on n=15,663 markets, 3.6× cost). Politics and economics are a level shift, not a slope— see F4.

F2 · Does the bias converge as the market runs out of time?

convergence check

Mean absolute deviation per time-to-close band. A price-discovery story says this must fall toward the close. It does for crypto and politics; it does the opposite for sports.



family	>60m	<2m	verdict
Crypto 15-min	—	—	n/a
Crypto hourly/daily	3.35	1.23	PASS
Sports game	4.91	6.84	FAIL
Sports props	3.38	9.20	FAIL
Politics/elections	4.06	1.01	PASS
Economics	—	—	n/a

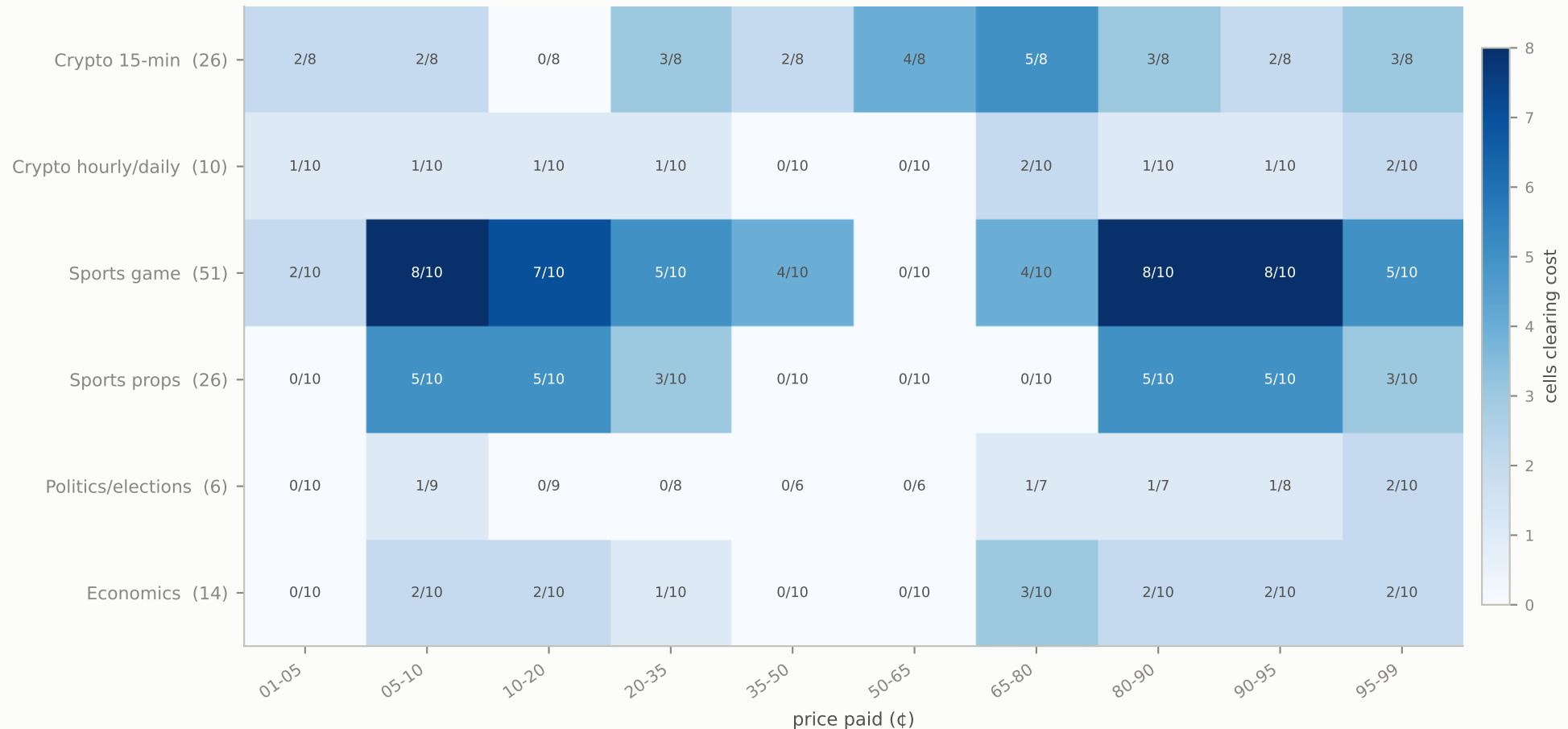
Two families have no comparison to make: crypto 15-min markets never live long enough to have a >60m band, and economics has almost no volume in the last minutes. The sports failures were chased down rather than waved away— settlement_ts is never earlier than close_time anywhere in the warehouse (median lag 3 min on sports, 6 s on crypto 15-min), so the anchor is clean. The <2m sports band is in-game trading, where a losing side's comeback ticket gets more expensive as the clock runs out. The spec's "must converge" premise simply does not hold for in-game markets; the sports time dimension stays suspended until the pre-game/in-game split (done in the maker replay follow-on).

F3 · Where the bias beats its own cost

net of cost

Each square is one family × price bucket. The number is how many of its cells (side × time band) showed a deviation larger than the round-trip cost with $n \geq 30$ and a confidence interval clear of zero, out of how many were measured.

Cells that clear their own round-trip cost, out of the cells measured



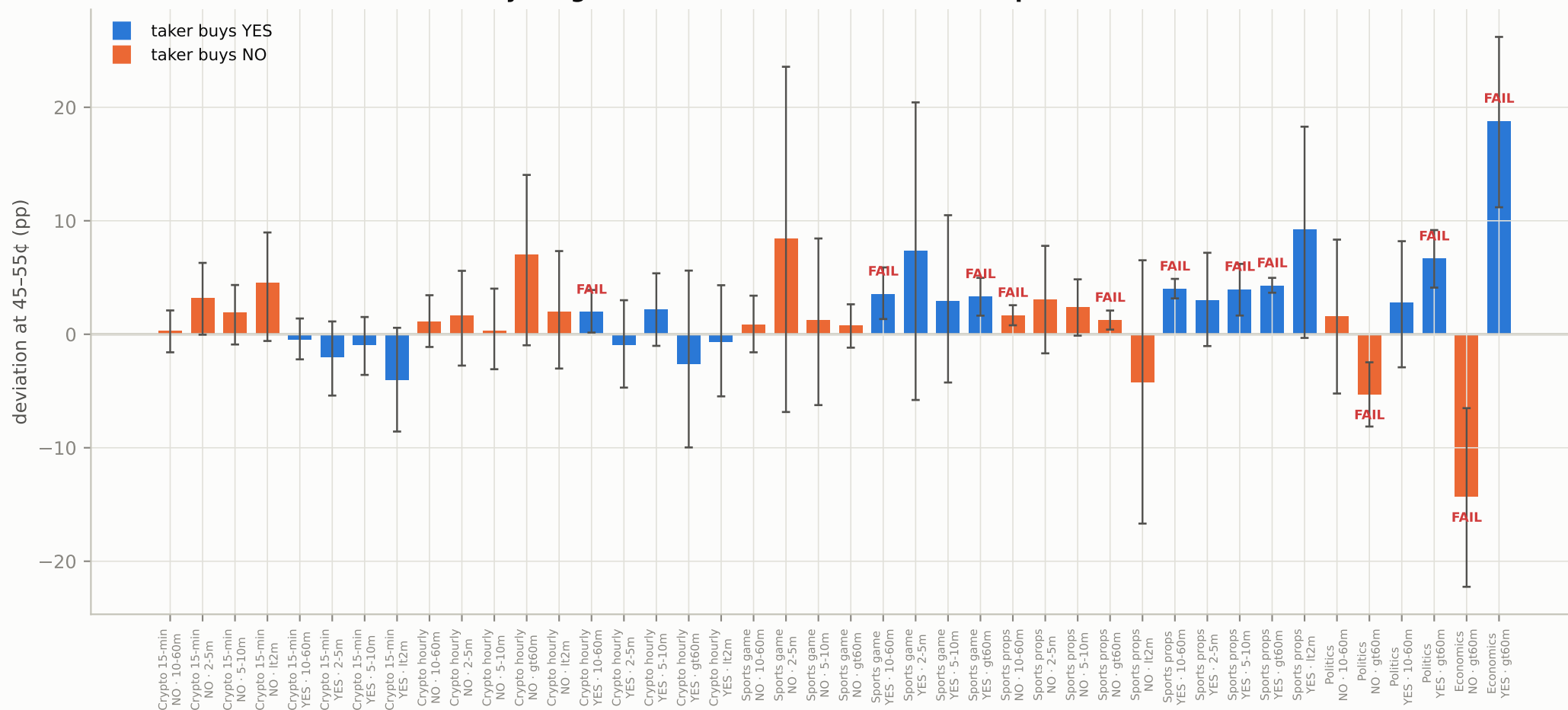
The number beside each family is its row total; 133 of 613 cells clear cost across the study. Sports game is the densest (51 of 110), sports props carries the largest volume. The 35–65¢ columns are almost empty, which is what a price-driven bias should look like— the deviation lives at the ends. Cost is the median in-cell spread plus the taker fee, so a cell that merely covers its own spread does not appear. This is a measurement, not a trade list: every position is held to settlement, no maker fee is charged, the fill is assumed at the volume-weighted taker price, and nothing here says the queue would ever have given us one. The maker replay follow-on answers those questions, and it kills half of this map.

F4 · The null-hypothesis cell

45-55¢ cell

The test that separates FLB from something else. At 45-55¢ a price-symmetric bias has nothing to say — both sides are near even money — so any deviation here is a different animal. Each bar is one family × side × time band.

Everything here should be zero. In econ and politics it is not.



Bars are 95% cluster-bootstrap intervals over markets; a cell FAILS when the interval excludes zero. Crypto passes across the board (1 of 19 cells grazes). Economics fails hardest — YES +18.8pp against NO -14.3pp, mirror-imaged, which is the signature of the market systematically over-pricing YES rather than over-pricing the longshot. Politics fails in the >60m band the same way. Same direction as the 07-18 audit's "optimism tax". Practical consequence: for these two families the price-bucket gradient on page 2 cannot be read as FLB until the directional component is netted out, and both families are in any case a fast-settling subset (68% of politics and 91% of elections volume never joined settlement inside the window).

Method, checks, and the three confounds that travel with these numbers

method

Definitions, fixed in code

Price is what the TAKER actually paid, and cells are built per taker side, so the YES and NO grids come from different flow and the complementarity check is a real test rather than an identity. The time anchor is `close_time` (it updates on early close). Cost is the median in-cell L1 spread plus the taker fee $7 \cdot p \cdot (1-p)$, with no closing leg and no settlement fee. The sample unit is the market; cells with $n < 30$ are greyed.

Complementarity — the check that the grid is measuring something real

Of 220 YES/NO mirror pairs, 150 (68.2%) have opposite signs; restricted to the 90 pairs where both sides deviate by at least 2pp, 84 (93.3%) do. The handful of same-sign violations all sit in the middle buckets, where both sides pay the spread— economically coherent, not a measurement error.

Three confounds that must be quoted with any number from this study

① Both crypto families drifted upward across the 16 days — a directional difference is not a bias. ② The sports >60m band mixes in-game prints, which is why the convergence check fails; the fix is the phase split, not a different anchor. ③ Politics (68% of volume), elections (91%) and economics (47%) largely never settled inside the window, so those families are a fast-settling selection, dominated by Mentions markets.

Gap events ($\geq 30\text{¢}$ jump between adjacent prints in one market)

Counted as a robustness read on whether the price series is continuous enough for the bucket assignment to mean anything. Median absolute jump is 37–42¢ in every family, i.e. the criterion is picking up genuine repricings rather than tick noise.

family	gap events	markets	median jump	criterion valid
crypto_15m	140	72	37¢	yes
crypto_hourly	2706	979	41¢	yes
econ	362	155	40¢	yes
politics_events	10844	1672	40¢	yes
sports_game	7726	1402	39¢	yes
sports_prop	73547	25312	42¢	yes